PROJECT REPORT

CHAPTER-1 : INTRODUCTION

1.1 Project Overview

The **India Agricultural Crop Production Analysis (1997–2021)** project is a data analytics and visualization project developed using Tableau. The project focuses on analyzing agricultural crop production trends in India over a period of 25 years, from 1997 to 2021.

The primary objective of this project is to provide comprehensive insights into agricultural production patterns by examining various factors such as:

- State-wise agricultural production
- Crop-wise production trends
- Seasonal crop distribution
- Year-wise production growth
- High and low yielding crops
- Comparative analysis of different agricultural regions

The project utilizes interactive dashboards, calculated fields, filters, KPIs, and storytelling features of Tableau to present meaningful insights. Three dashboards and six story points were developed to enable users to explore agricultural trends efficiently.

The analysis is designed to support various stakeholders, including:

- **Farmers**, to understand crop performance and production patterns.
- **Government policymakers**, to formulate agricultural policies and resource allocation strategies.
- **Researchers and analysts**, to identify trends and conduct further agricultural studies.

The project demonstrates how data analytics and visualization can assist in informed decision-making and contribute to the improvement of agricultural productivity and planning.

1.2 Purpose

The purpose of this project is:

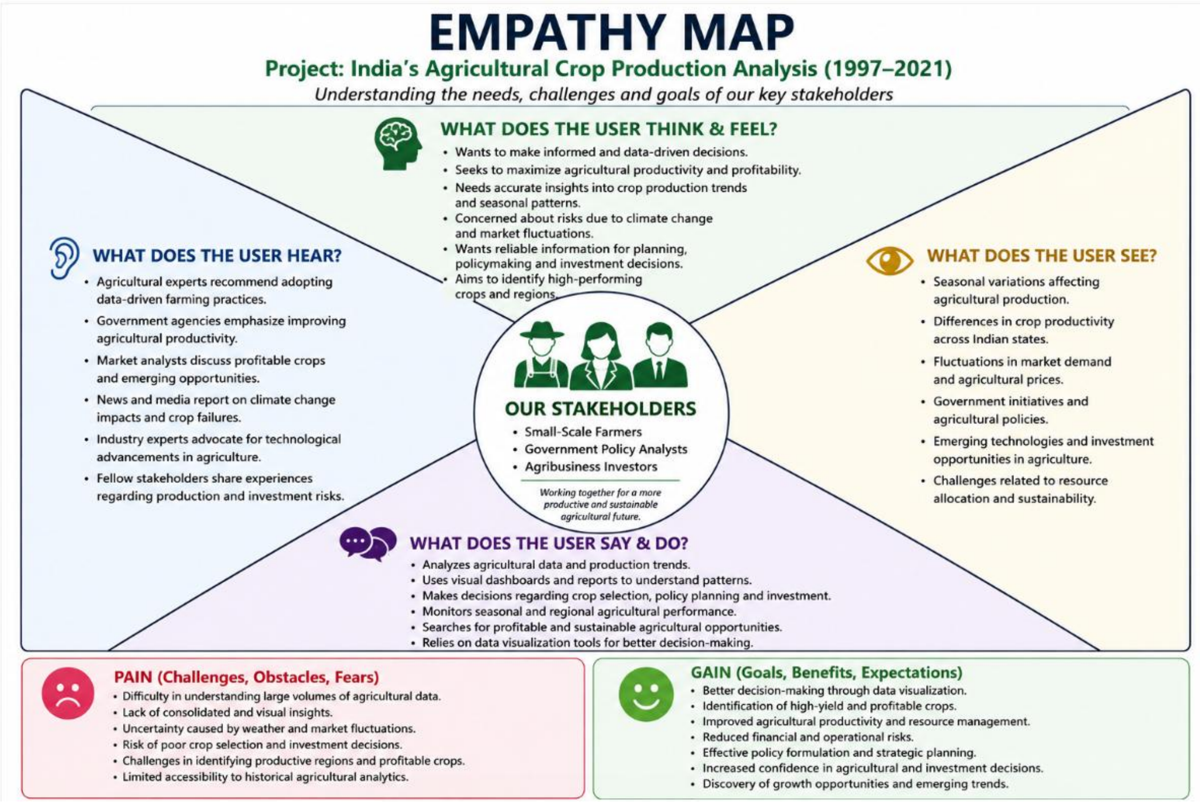
- To analyze crop production trends in India.
- To identify high-performing states and crops.
- To study seasonal agricultural patterns.
- To develop interactive Tableau dashboards.
- To assist stakeholders in making data-driven decisions.

CHAPTER-2 : IDEATION PHASE

2.1 Problem Statement

Farmers, agricultural policymakers, and researchers struggle to analyze large agricultural datasets due to fragmented information and lack of visual insights. This project aims to provide a comprehensive analytical dashboard that enables effective decision-making through interactive visualizations.

2.2 Empathy Map Canvas



2.3 Brainstorming

Team members brainstormed ideas and identified the major challenges:

- State-wise production analysis
- Crop performance analysis
- Seasonal trend analysis
- Yield comparison
- Dashboard storytelling

Team Members:

Member	Responsibility
Garima Rao	Data Collection
Mohan Singh Shekhawat	Visualization Design
Yusuf Khan	Dashboard Development
Komal Goyal	Story Creation & Documentation
Durlabh Kaushal	Testing & Deployment

CHAPTER-3: REQUIREMENT ANALYSIS

3.1 Customer Journey Map

3.2 Solution Requirement

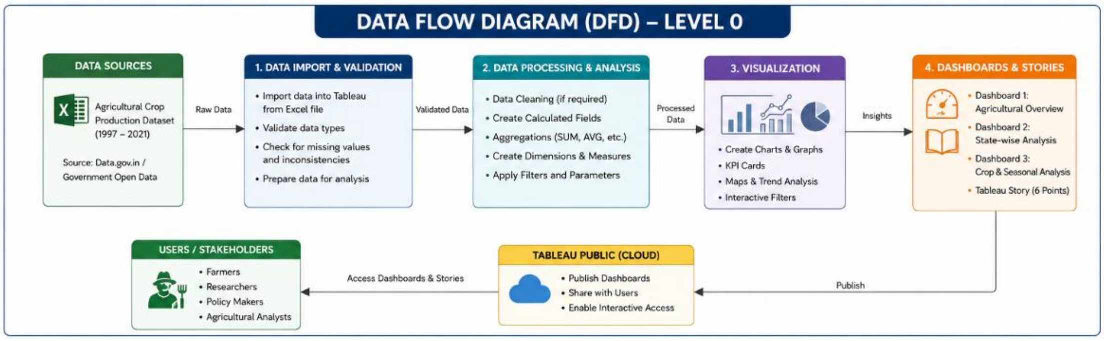
Functional Requirements are as follows:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import & Processing	Import dataset, validate data, create calculated fields
FR-2	Dashboard Development	Agricultural Overview, State Analysis, Crop Analysis dashboards
FR-3	Interactive Visualization	Filters, charts, maps, KPI cards, drill-down analysis
FR-4	Story & Publishing	Tableau Story creation, insights generation, Tableau Public publishing

Non Functional Requirements are as follows:

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The dashboards should provide an intuitive and user-friendly interface with interactive filters, charts, and visualizations that can be easily understood by farmers, researchers, and policymakers.
NFR-2	**Security**	The project uses publicly available agricultural data and is published through Tableau Public, ensuring safe access without requiring user authentication or sensitive data handling.
NFR-3	**Reliability**	The system should consistently display accurate agricultural insights and maintain data integrity across all dashboards and story visualizations.
NFR-4	**Performance**	The dashboards should load efficiently and respond quickly to filter selections, drill-down operations, and visualization rendering.
NFR-5	**Availability**	The dashboards and stories published on Tableau Public should remain accessible to users anytime through an internet connection.
NFR-6	**Scalability**	The solution should support the addition of future agricultural datasets, new visualizations, and extended analysis without major redesign.

3.3 Data Flow Diagram



3.4 Technology Stack

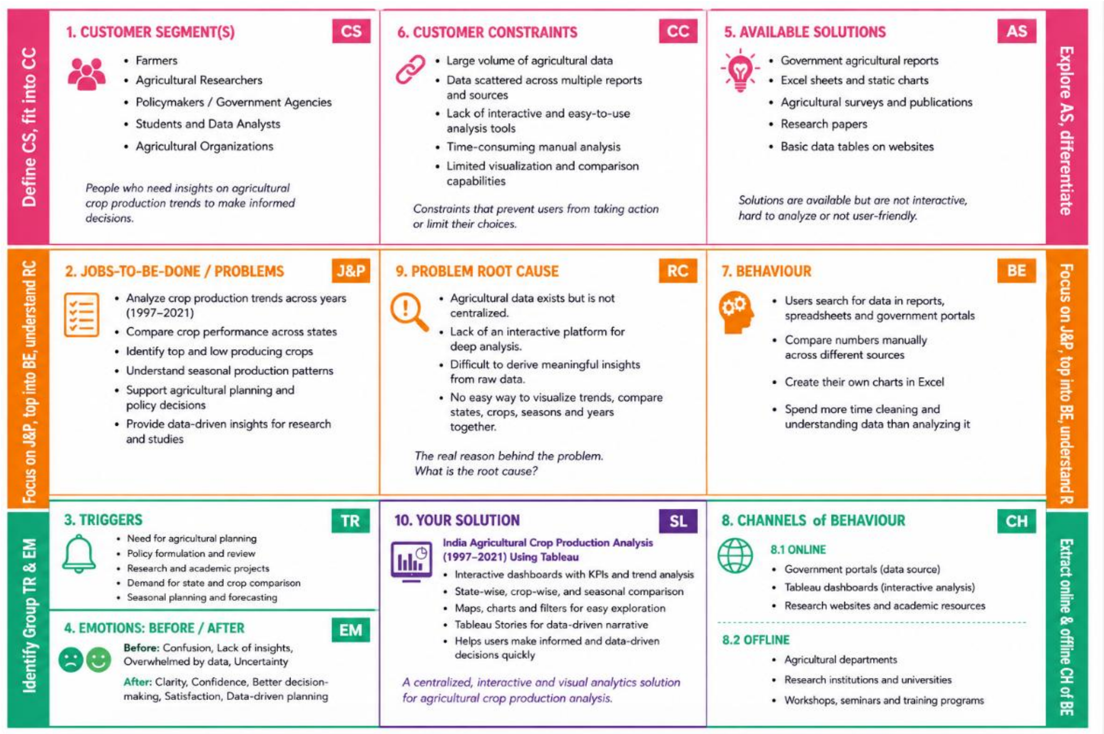
TECHNOLOGY STACK

The following technologies and tools were used to design, develop, analyze and deploy the project.

TECHNOLOGY / TOOL	PURPOSE	KEY FEATURES / USAGE
tableau Tableau	Data Visualization and Dashboard Development	• Interactive Dashboards • Data Exploration • Storytelling • KPI & Calculated Fields • Filters & Parameters
Microsoft Excel	Data Storage and Initial Data Preparation	• Data Collection • Data Structuring • Basic Cleaning • Data Validation
tableau public Tableau Public	Publishing and Sharing Dashboards	• Publish Workbooks • Share Public Link • Make Visualizations Accessible
GitHub	Version Control and Collaboration	• Store Project Files • Track Changes • Team Collaboration
Microsoft Word	Documentation and Reporting	• Project Documentation • Report Preparation • Format & Structure
Internet / Web	Research and Data Collection	• Data Research • Validating Information • Reference and Insights

CHAPTER-4 : PROJECT DESIGN

4.1 Problem Solution Fit



4.2 Proposed Solution

The proposed system analyzes agricultural data through:

- Dashboard 1: Overview Analysis
- Dashboard 2: State Analysis
- Dashboard 3: Crop & Seasonal Analysis
- Story Points

4.3 Solution Architecture

Agricultural Dataset

↓

Data Validation

↓

Tableau

↓

KPI Calculation

↓

Visualizations

↓

Dashboards

↓

Story Analysis

↓

Decision Support

CHAPTER-5 : PROJECT PLANNING & SCHEDULING

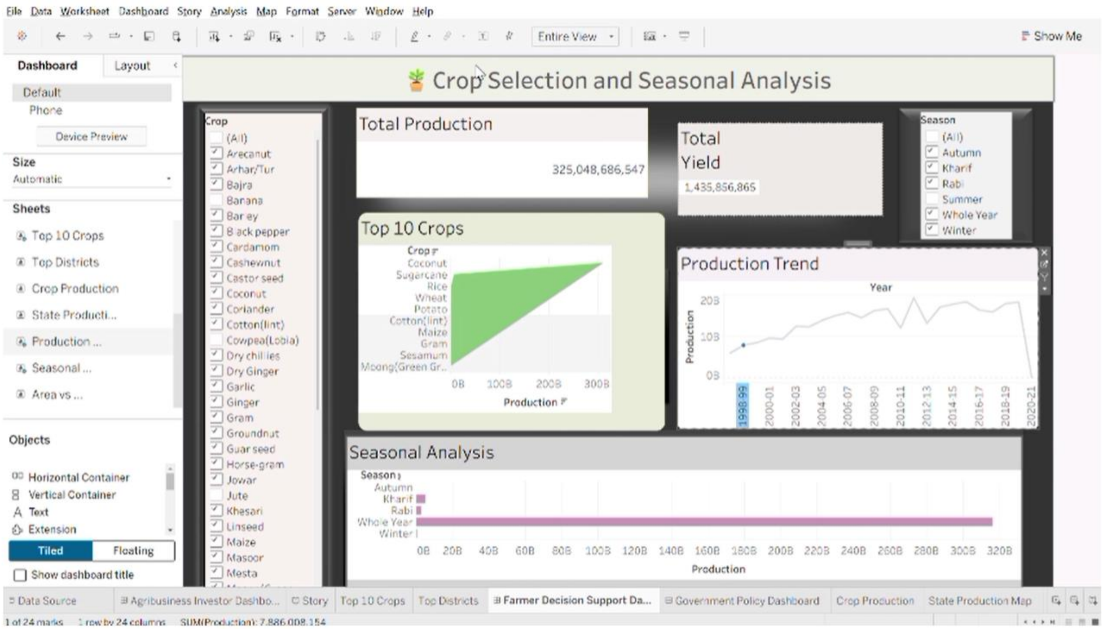
Activity	**Duration**
Project Identification	1 day
Data Collection	1 day
Dashboard Design	1 day
Story Development	1 day
Testing	1 day
Deployment	1 day

CHAPTER-6 : FUNCTIONAL & PERFORMANCE TESTING

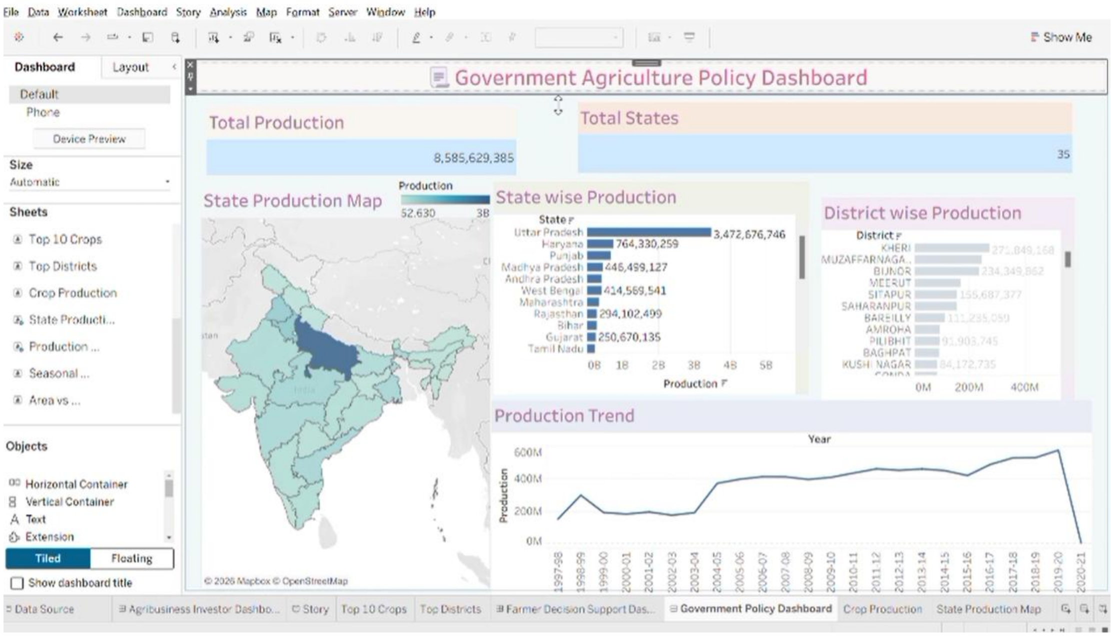
Parameter	**Result**
Data Rendering	Successful
Data Validation	Successful
Filters	Working
Calculated Fields	Working
Dashboard Design	3 Dashboards
Story Design	6 Story Points

CHAPTER-7 : RESULTS

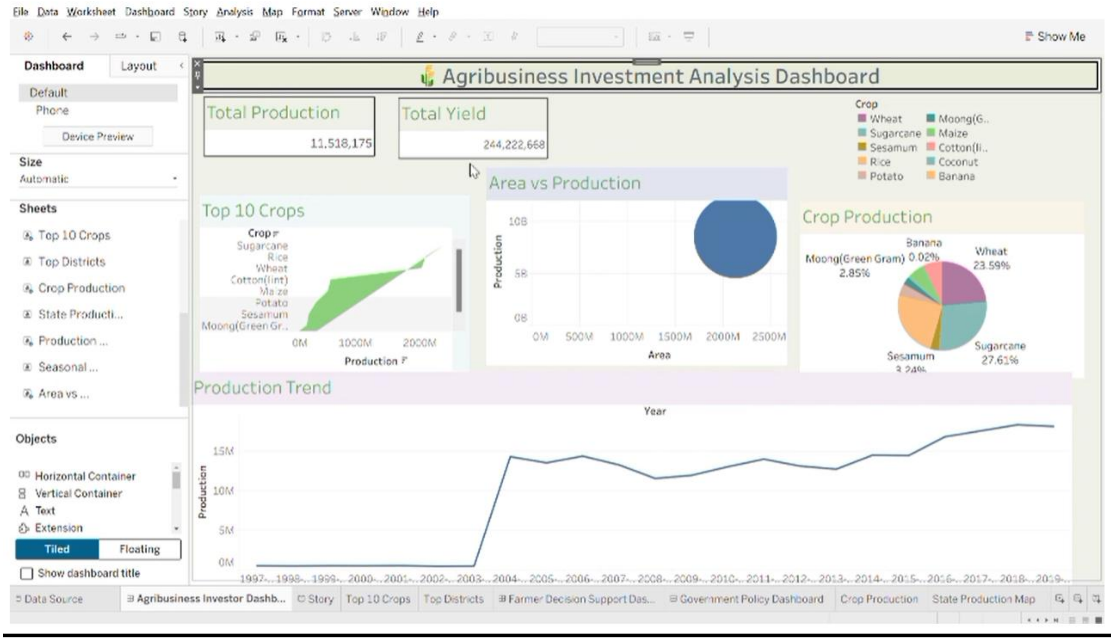
<u>Dashboard 1</u>



Dashboard 2



Dashboard 3



CHAPTER-8 : ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboards
- Easy analysis
- Better decision-making
- User-friendly visualization
- Dynamic filtering

Disadvantages

- Depends on dataset quality
- Requires Tableau knowledge
- Limited predictive capability

CHAPTER-9 : CONCLUSION

This project successfully analyzed India's agricultural crop production from 1997 to 2021 using Tableau. Interactive dashboards and story visualizations provided meaningful insights regarding crop production trends, seasonal patterns, and state-wise agricultural performance.

CHAPTER-10 : FUTURE SCOPE

Future enhancements include:

- Machine Learning prediction
- Real-time agricultural data integration
- Weather forecasting analysis
- AI-powered recommendations
- Mobile dashboard development

CHAPTER 11 : APPENDIX

- **Tableau Workbook(Dataset)**
- **Tableau Public Link**
- **Github Repository Link**
- **Project Demo**